Econ 30303 StudyGuide

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Table of contents

Introduction 3

1 Chapter 8: Production 4
- 1.1 Main Summary . . . 4
  - 1.1.1 Key Definitions . . . 5
  - 1.1.2 Key Formulas . . . 5
  - 1.1.3 Important Relationships . . . 6
  - 1.1.4 Common Mistakes to Avoid . . . 6
- 1.2 Practice Questions . . . 6
  - 1.2.1 Part A: Multiple Choice . . . 6
  - 1.2.2 Part B: Calculation Questions . . . 8

2 Chapter 13: The Costs of Production 11
- 2.1 1. Economic vs. Accounting Costs . . . 11
- 2.2 2. Production Function and Short-Run Costs . . . 12
- 2.3 3. Key Relationships Between the Curves . . . 12
- 2.4 4. Long-Run Costs . . . 13
- 2.5 5. Common Mistakes to Avoid . . . 13
- 2.6 Practice Questions . . . 13
  - 2.6.1 Part A: Multiple Choice . . . 13
  - 2.6.2 Part B: Calculation Questions . . . 17

Appendices 20
- A.1 Main Summary . . . 20

Introduction

The begining is the most important part of a journey!

- Syllabus
- Course Calendar
- Formula Sheet

Continue to the next page to see exam 1 study guide page!

1 Chapter 8: Production

Study Guide and Practice (Intermediate Micro)

1.1 Main Summary

The chapter explains how a firm turns inputs into output. Its main ideas are:

1. **The production function** links inputs (capital, labor, land, technology, and so on) to the maximum output they can produce.
2. **Short run vs. long run:** in the short run at least one input is fixed. In the long run every input can change.
3. **Diminishing returns:** in the short run, adding more of one variable input to fixed inputs eventually adds less and less output.
4. **Total, average, and marginal product** are three linked ways to describe output per unit of a variable input.
5. **Efficient allocation:** a divisible input should go where its marginal product is highest, which means its marginal product is equal across activities at the optimum.
6. **Isoquants** show all input combinations that produce the same output. The **marginal rate of technical substitution** (MRTS) measures how much of one input you can give up when you add a little of the other, while holding output constant.
7. **Returns to scale** describe what happens when all inputs grow proportionally: output can rise more than proportionally (increasing), proportionally (constant), or less than proportionally (decreasing).
8. **Technological progress** shifts the production function upward, so the same inputs produce more output.

The appendix extends the chapter with specific functional forms: the Cobb-Douglas function, the Leontief (fixed-proportions) function, and the "production mountain" view of isoquant maps.

1.1.1 Key Definitions

- **Production function:** the relationship between inputs and the maximum output they can produce.
- **Long run:** the period long enough for all inputs to be varied.
- **Short run:** the period during which at least one input cannot be varied.
- **Variable input:** an input whose quantity can change in the short run.
- **Fixed input:** an input whose quantity cannot change in the short run.
- **Total product (TP):** total output as a function of the variable input.
- **Marginal product (MP):** the change in total output from one more unit of the variable input.
- **Average product (AP):** total output divided by the quantity of the variable input.
- **Isoquant:** the set of all input combinations that produce the same level of output.
- **MRTS:** the rate at which one input can replace another while output stays the same.
- **Returns to scale:** how output responds when all inputs are scaled up by the same proportion.

1.1.2 Key Formulas

With labor L as the variable input and capital K fixed in the short run:

$$AP_L = \frac{Q}{L}, \qquad MP_L = \frac{\Delta Q}{\Delta L}$$

For two inputs, the marginal rate of technical substitution is

$$MRTS_{LK} = -\frac{\Delta K}{\Delta L}\bigg|_{Q \text{ fixed}} = \frac{MP_L}{MP_K}$$

Returns to scale are determined by multiplying every input by a constant $c > 1$ in $Q = F(K, L)$:

$$F(cK, cL) \begin{cases} > cF(K, L) & \text{increasing returns} \\ = cF(K, L) & \text{constant returns} \\ < cF(K, L) & \text{decreasing returns} \end{cases}$$

For the Cobb-Douglas function $Q = AK^{\alpha}L^{\beta}$, the sum $\alpha + \beta$ tells you the returns to scale: greater than 1 is increasing, equal to 1 is constant, and less than 1 is decreasing.

For the Leontief function $Q = \min(aK, bL)$, inputs must be used in fixed proportions.

1.1.3 Important Relationships

- When $MP_L > AP_L$, AP_L is **rising**. When $MP_L < AP_L$, AP_L is **falling**.
- MP_L intersects AP_L at the **maximum of** AP_L.
- Total product is maximized where $MP_L = 0$.
- Along a single isoquant, MRTS falls as you move down and to the right for a convex isoquant. This is the "diminishing MRTS" property.
- **Perfect substitutes:** the isoquant is a straight line, and MRTS is constant.
- **Perfect complements:** the isoquant is L-shaped, with a corner at the fixed input proportion.

1.1.4 Common Mistakes to Avoid

- Calling the law of diminishing returns a claim that output falls. Output can still rise; the increase is just smaller each time.
- Confusing returns to scale (all inputs change) with diminishing returns (one input changes).
- Assuming a higher average product means a higher marginal product. When AP is rising, MP is above AP, but MP itself can be falling.
- Reading the allocation rule as "put the input where it is most productive on average." The rule uses marginal product.

1.2 Practice Questions

1.2.1 Part A: Multiple Choice

Question 1. A factory has two machines that are fixed for the year. It can hire more or fewer workers at any time. Which statement best describes the situation?

- A. The firm is in the long run because it can change its labor input.
- B. The firm is in the short run because at least one input (the machines) cannot change.
- C. The firm is in the long run because it can change its output.
- D. The firm is in the short run only if it has a single worker.

Answer Key

Answer: B. The short run is defined by at least one fixed input. Here the machines are fixed, so the firm is in the short run. Labor can change, but that does not make the period long run.

Question 2. On a graph of average product of labor (AP) and marginal product of labor (MP) against labor, which statement is correct?

- A. AP reaches its maximum where MP is at its maximum.
- B. AP reaches its maximum where MP crosses AP.
- C. AP reaches its maximum where total product is at its maximum.
- D. AP reaches its maximum where MP equals zero.

Answer Key

Answer: B. When MP is above AP, AP is rising. When MP is below AP, AP is falling. So AP peaks where the two curves cross. Total product peaks where MP = 0, which is a different point (option C and D confuse the two).

Question 3. A manager has 10 workers to split between two production lines. At the current split, the marginal product of a worker on Line A is 12 and on Line B is 18. Which action best improves output, assuming workers can be moved one at a time?

- A. Move a worker from Line A to Line B.
- B. Move a worker from Line B to Line A.
- C. Keep the current split, because total output is already at its maximum.
- D. Move workers so that the average products on both lines are equal.

Answer Key

Answer: A. Moving a worker from a line where MP is 12 to a line where MP is 18 adds more output than it removes. The allocation is efficient only when MP is equal across lines. Option D uses average products, which is the wrong comparison.

Question 4. A new production technique lets a firm make 50 more units of output from the same amount of labor and capital. What does this represent?

- A. A movement along the existing production function.
- B. An upward shift of the production function (technological progress).
- C. A move from the long run to the short run.
- D. Increasing returns to scale.

Answer Key

Answer: B. The same inputs now produce more output, so the production function has shifted up. A movement along a fixed function would require changing inputs. Increasing returns refer to scaling all inputs, which did not happen.

Question 5. Which statement about isoquants is correct?

- A. Two isoquants for the same firm can cross, because a firm can produce different outputs with the same inputs.
- B. Isoquants for higher output levels lie closer to the origin.

- C. For perfect substitutes, the isoquant is L-shaped, and for perfect complements, it is a straight line.
- D. For a convex isoquant, the MRTS falls as one moves down the curve, because the input being given up becomes more valuable.

Answer Key

Answer: D. Isoquants for different output levels do not cross, and higher outputs lie farther from the origin, so A and B are false. Perfect substitutes give straight-line isoquants and perfect complements give L-shaped ones, so C has the shapes reversed. The statement in D describes the diminishing MRTS.

Question 6. A firm doubles all of its inputs and its output rises from 100 units to 180 units. Which statement is correct?

- A. The firm has constant returns to scale.
- B. The firm has increasing returns to scale.
- C. The firm has decreasing returns to scale.
- D. The firm is in the short run, so returns to scale do not apply.

Answer Key

Answer: C. Doubling inputs should double output under constant returns. Output rose by only 80 percent (to 180 from 100, which is 1.8 times the original), so output rose less than proportionally. That is decreasing returns to scale. Returns to scale apply when all inputs change, which is the case here, so D is wrong.

1.2.2 Part B: Calculation Questions

Question 7. A firm uses labor L as its only variable input, with capital fixed. The total product of labor is shown in the R table below.

Table 1.1

L	TP	MP_L	AP_L
0	0	NA	NA
1	12	12	12.0
2	28	16	14.0
3	42	14	14.0
4	52	10	13.0
5	58	6	11.6
6	60	2	10.0

(a) Find the labor level at which AP_L is at its maximum, and its value.

(b) Find the labor level at which total product is at its maximum, and explain why it is different from the labor level where $MP_L = 0$.
(c) Is the law of diminishing returns in effect over the range shown? Explain.

Answer Key

(a) AP_L is 12 at $L = 1$, 14 at $L = 2$, 14 at $L = 3$, 13 at $L = 4$, and lower after that. The maximum is 14, reached at $L = 2$ and $L = 3$ (tied). At $L = 3$, $MP_L = 14$ equals AP_L, which is consistent with the rule that MP crosses AP at AP's maximum.

(b) Total product is highest at the largest labor level shown, $L = 6$, where $TP = 60$. But MP_L is still positive there ($MP_6 = 2$), so output is still rising, just more slowly. Total product is maximized where $MP_L = 0$, which lies beyond $L = 6$ in this data. The table shows the maximum only because the data stop at $L = 6$.

(c) Yes. MP_L rises from 12 to 16 (up to $L = 2$) and then falls from 16 to 14, 10, 6, 2. After $L = 2$, each additional worker adds less output than the one before, which is diminishing returns.

Question 8. A firm's production function is $Q = K^{0.5}L^{0.5}$.

(a) Find MP_L and MP_K.
(b) Find the marginal rate of technical substitution, $MRTS_{LK}$, and explain what it means at $K = 16$, $L = 4$.
(c) What are the returns to scale? Justify your answer.

Answer Key

(a)

$$MP_L = \frac{\partial Q}{\partial L} = 0.5\,K^{0.5}L^{-0.5}, \qquad MP_K = \frac{\partial Q}{\partial K} = 0.5\,K^{-0.5}L^{0.5}$$

(b)

$$MRTS_{LK} = \frac{MP_L}{MP_K} = \frac{K}{L}$$

At $K = 16$ and $L = 4$, $MRTS_{LK} = 16/4 = 4$. This means that, at that point, the firm can give up 1 unit of capital and replace it with 4 units of labor while keeping output unchanged. Equivalently, one unit of labor replaces 0.25 units of capital.

(c) Constant returns to scale. Replace K and L with cK and cL:

$$Q' = (cK)^{0.5}(cL)^{0.5} = c^{0.5+0.5}K^{0.5}L^{0.5} = cQ$$

The exponents sum to $\alpha + \beta = 0.5 + 0.5 = 1$, so output scales exactly with inputs.

Question 9. A Leontief production function is $Q = \min(2K,\ 3L)$.

(a) How many units of labor are needed with 6 units of capital to produce 12 units of output?
(b) If the firm has 6 units of capital and 6 units of labor, what is its output? Which input is in excess?
(c) What is the MRTS along any isoquant of this function?

Answer Key

(a) With $K = 6$, $2K = 12$. To produce $Q = 12$ you need $3L \geq 12$, so $L = 4$. With exactly this combination, both inputs are fully used.

(b) $Q = \min(2 \times 6,\ 3 \times 6) = \min(12, 18) = 12$. Capital is the binding input (its limit is 12), and labor has 6 units but only 4 are needed. Labor is in excess by 2 units.

(c) The isoquant is L-shaped, with a corner where $2K = 3L$. Along the vertical and horizontal segments, one input can change while output stays the same. The MRTS is 0 or undefined there, which means the inputs are not substitutes. At the corner, the inputs must be used together in the fixed proportion $K/L = 3/2$. So MRTS does not describe substitution for perfect complements.

Question 10. A firm has 10 workers to split between two plots of land, A and B. The marginal product of labor on each plot is

$$MP_A = 20 - 2L_A, \qquad MP_B = 12 - L_B$$

where $L_A + L_B = 10$. Find the allocation that maximizes total output, and the marginal product on each plot at that allocation.

Answer Key

Efficient allocation requires equal marginal products:

$$20 - 2L_A = 12 - L_B$$

Substitute $L_B = 10 - L_A$:

$$20 - 2L_A = 12 - (10 - L_A) = 2 + L_A \quad \Rightarrow \quad 18 = 3L_A \quad \Rightarrow \quad L_A = 6$$

So $L_B = 4$. Check: $MP_A = 20 - 12 = 8$, and $MP_B = 12 - 4 = 8$. The marginal products are equal, so no reallocation can add output.

2 Chapter 13: The Costs of Production

Intermediate Micro Study Notes and Practice

2.1 1. Economic vs. Accounting Costs

- **Explicit costs** are payments made to outsiders: wages, rent, materials, interest on borrowed funds.
- **Implicit costs** are the value of resources the owner already controls and does not pay for directly, such as the owner's forgone salary, or interest the owner could have earned on savings used in the business.

Economic cost, accounting profit, and economic profit are related as follows:

$$\text{Economic cost} = \text{Explicit costs} + \text{Implicit costs}$$

$$\pi_{\text{accounting}} = TR - \text{Explicit costs}$$

$$\pi_{\text{economic}} = TR - (\text{Explicit costs} + \text{Implicit costs}) = \pi_{\text{accounting}} - \text{Implicit costs}$$

- An economic profit of zero is not a failure. It means the owner earns exactly what the resources could earn in their next-best use (a normal return).
- **Sunk costs** are costs already committed that cannot be recovered. They should not affect forward-looking decisions.

2.2 2. Production Function and Short-Run Costs

- A **production function** links inputs to output, for example $Q = f(L, K)$.
- In the **short run**, at least one input is fixed (for example, the factory). In the **long run**, every input can be changed.
- **Marginal product of labor:** $MP_L = \dfrac{\Delta Q}{\Delta L}$
- **Diminishing marginal product:** after some point, each added worker adds less output than the one before, given fixed capital.

Cost definitions, with FC = fixed cost and VC = variable cost:

$$TC = FC + VC$$

$$AFC = \frac{FC}{Q}, \qquad AVC = \frac{VC}{Q}, \qquad ATC = \frac{TC}{Q} = AFC + AVC$$

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta VC}{\Delta Q}$$

- **Link between MC and MP:** if labor is the only variable input and the wage is w, then

$$MC = \frac{w}{MP_L}$$

  When MP_L rises, MC falls. When MP_L falls, MC rises. This is why the marginal cost curve rises in the short run: it reflects diminishing marginal product, with input prices held constant.

- AFC always falls as output rises, because fixed cost is spread over more units.
- AVC, ATC, and MC are typically U-shaped.

2.3 3. Key Relationships Between the Curves

- If $MC < ATC$, ATC is falling. If $MC > ATC$, ATC is rising.
- Therefore, **MC crosses ATC at the minimum of ATC**.
- Similarly, **MC crosses AVC at the minimum of AVC**.
- The minimum of ATC occurs at a higher output than the minimum of AVC, because ATC also includes AFC, which is still falling at that point.
- A change in fixed cost shifts AFC and ATC, but does **not** change MC or AVC.

2.4 4. Long-Run Costs

- In the long run, all inputs are variable, so the firm can choose its best plant size.
- The **long-run average total cost (LRATC)** curve is the envelope of the short-run ATC curves: it touches each SRATC curve at one point.
- **Economies of scale:** LRATC falls as output rises (specialization, division of labor, cheaper bulk inputs).
- **Constant returns to scale:** LRATC is flat.
- **Diseconomies of scale:** LRATC rises as output rises (coordination problems, management layers, bureaucracy).
- **Minimum efficient scale (MES):** the smallest output at which LRATC reaches its minimum.

2.5 5. Common Mistakes to Avoid

- Treating accounting profit as economic profit. Always subtract the implicit cost of the owner's time and money.
- Including sunk costs in a decision about the future.
- Saying MC rises "because costs rise." MC rises because marginal product falls (with input prices held constant).
- Reading the ATC minimum from the MC curve without checking that MC crosses ATC there.
- Thinking that a fixed-cost increase raises MC. It does not.

2.6 Practice Questions

2.6.1 Part A: Multiple Choice

Question 1. A bakery's weekly short-run data are shown below.

Output (loaves)	0	10	20	30	40	50
Total cost ($)	600	900	1,100	1,350	1,650	2,150

Which statement is correct?

- A. ATC is lowest at 20 loaves, because MC is lowest over the interval from 10 to 20.

- B. ATC is lowest at 40 loaves (about $41.25 per loaf), and ATC rises to $43 at 50 loaves.
- C. ATC is lowest at 30 loaves, because MC equals ATC at that output.
- D. Fixed cost is $1,100 per week.

Answer Key

Answer: B.

ATC at each output: 10 loaves → 90, 20 → 55, 30 → 45, 40 → 41.25, 50 → 43. ATC is lowest at 40 loaves. MC over each interval is 30, 20, 25, 30, 50. MC is lowest between 10 and 20, but that does not determine the ATC minimum. Fixed cost is $600 (the TC at zero output), so D is wrong.

Question 2. A small business owner's accounting profit for the year is $80,000. She could have earned $50,000 working for an employer, and the $400,000 she has invested in the business could have earned $30,000 in interest elsewhere. Which statement is correct?

- A. Her economic profit is $80,000, because the business is profitable by accounting standards.
- B. Her economic profit is zero, because implicit costs total $80,000.
- C. Her economic profit is −$30,000, because the implicit costs total $110,000.
- D. Her economic profit is $30,000, because only the interest on capital is an implicit cost.

Answer Key

Answer: B.

Implicit costs = forgone salary ($50,000) + forgone interest ($30,000) = $80,000. Economic profit = accounting profit − implicit costs = $80,000 − $80,000 = 0. The owner earns a normal return: the business is just as good as her next-best use of her time and money.

Question 3. A firm hires workers at a fixed wage. The marginal product of labor rises from the first worker to the third worker and then falls for each additional worker. Which statement is correct?

- A. Marginal cost falls as long as marginal product rises, then rises once marginal product begins to fall.
- B. Marginal cost rises from the first worker onward, because the wage is constant.
- C. Marginal cost is constant, because the wage is constant.
- D. Marginal cost falls indefinitely, because more workers always lower the cost per unit.

Answer Key

Answer: A.

Because $MC = w/MP_L$ with a fixed wage, MC moves opposite to MP. MC falls while MP rises and rises once MP falls. The wage does not change, so it cannot explain the rise in MC.

Question 4. A firm's long-run average total cost falls from an output of 100 units to 300 units, stays flat from 300 to 500 units, and rises beyond 500 units. Which statement is correct?

- A. Diseconomies of scale begin at 300 units.
- B. The minimum efficient scale is 300 units.
- C. There are constant returns to scale below 300 units.
- D. The minimum efficient scale is 500 units, where the curve starts to rise.

Answer Key

Answer: B.

Economies of scale operate from 100 to 300 units. Between 300 and 500 units LRATC is flat (constant returns to scale). Diseconomies begin above 500 units. The minimum efficient scale is the smallest output that reaches the minimum LRATC, which is 300 units.

Question 5. A firm in a competitive market has a minimum AVC of \$14 (at 8 units) and a minimum ATC of \$20 (at 12 units). The market price is \$12. In the short run, which is correct?

- A. The firm should keep producing at 12 units, because price is below minimum ATC but the firm can still cover fixed costs.
- B. The firm should shut down, because price is below minimum AVC, so it cannot cover variable costs.
- C. The firm should produce 8 units, because that is where AVC is at its minimum.
- D. The firm should expand output to reach the lowest ATC at 12 units.

Answer Key

Answer: B.

In the short run, a firm should keep operating only if price covers AVC at its chosen output. With price of \$12 below the lowest AVC of \$14, no output level covers variable costs, so the firm shuts down and loses only its fixed cost. The ATC comparison is not the short-run shutdown test; A and D reflect that mistake.

Question 6. At an output of 150 units, a firm's ATC is \$24 and ATC is falling as output rises. Which must be true at 150 units?

- A. MC is greater than \$24.
- B. MC is less than \$24.
- C. MC equals \$24.
- D. AFC is rising.

Answer Key

Answer: B.

ATC falls only when MC is below ATC (the new unit pulls the average down). So MC $<$ \$24 at 150 units. AFC always falls as output rises, so D is wrong.

Question 7. A firm spent \$2 million last year on specialized equipment that has no resale value. It is now deciding whether to keep operating. Which is correct?

- A. The \$2 million should be included in the decision, because it is a cost the firm has already incurred.
- B. The \$2 million should be excluded, because it is a sunk cost; only future revenues and avoidable costs should be compared.
- C. The \$2 million should be divided over future output to find a higher ATC that justifies shutting down.
- D. The \$2 million should be counted as an implicit cost that reduces the firm's accounting profit.

Answer Key

Answer: B.

Sunk costs cannot be recovered whether the firm continues or stops, so they should not affect the forward-looking decision. Option D is wrong because the cost is explicit (it was paid), and it is already a past cost, not a forgone opportunity.

Question 8. A firm's monthly rent rises by \$200, with no change in output or variable inputs. Which is correct?

- A. MC rises by \$200 / Q, and AVC rises.
- B. AFC and ATC both rise, while MC and AVC are unchanged.
- C. MC and ATC both rise, while AFC falls.
- D. Only AVC rises, because rent is a variable cost.

Answer Key

Answer: B.

Rent is a fixed cost. A higher FC raises AFC (by \$200 divided by Q) and therefore ATC, but it does not change VC, AVC, or MC.

2.6.2 Part B: Calculation Questions

Question 9. A firm has fixed cost of \$50. Its total cost at outputs Q = 1 to 5 is given in the R table below. The table is generated from the total cost values, so you can check every entry yourself.

Table 2.1

Q	VC	AFC	AVC	ATC	MC
1	30	50.00	30.0	80.00	30
2	50	25.00	25.0	50.00	20
3	75	16.67	25.0	41.67	25
4	110	12.50	27.5	40.00	35
5	155	10.00	31.0	41.00	45

(a) Using the table, find the output at which ATC is at its minimum, and its value.
(b) Verify that $ATC = AFC + AVC$ at $Q = 3$.

Answer Key

(a) ATC is lowest at $Q = 4$, where $ATC = \$40.00$.

(b) At $Q = 3$: $AFC + AVC = 16.67 + 25.00 = 41.67 = ATC = 125/3 = 41.67$.

Question 10. A firm has fixed cost of \$500 and hires labor at \$100 per day. The R table below shows output for 0 to 5 workers and the resulting marginal product and marginal cost.

Table 2.2

Workers (L)	Output (Q)	MP_L	MC (\$)
1	10	10	10.00
2	26	16	6.25
3	36	10	10.00
4	40	4	25.00
5	42	2	50.00

(a) Using the table, describe how MC changes as MP changes.
(b) At 5 workers, compute AVC and ATC.

Answer Key

(a) MC = w/MP_L. MC falls from worker 1 to worker 2 while MP rises, then rises from worker 3 onward once MP falls.

(b) At 5 workers: $VC = 5 \times 100 = \$500$ and $TC = 500 + 500 = \$1{,}000$. $AVC = 500/42 \approx \$11.90$. $ATC = 1{,}000/42 \approx \$23.81$.

Question 11. A firm has the cost function

$$TC = 200 + 10Q + 0.5Q^2$$

The R table below evaluates ATC and MC for outputs from 10 to 30 units.

Table 2.3

Q	TC	ATC	MC
10	350.0	35.00	20
15	462.5	30.83	25
20	600.0	30.00	30
25	762.5	30.50	35
30	950.0	31.67	40

(a) Find expressions for FC, VC, AVC, ATC, and MC.
(b) Compute TC, AVC, ATC, and MC at $Q = 20$.
(c) Find the output that minimizes ATC, and the minimum ATC value.

Answer Key

(a)

$$FC = 200, \qquad VC = 10Q + 0.5Q^2$$

$$AVC = 10 + 0.5Q, \qquad ATC = \frac{200}{Q} + 10 + 0.5Q, \qquad MC = \frac{dTC}{dQ} = 10 + Q$$

(b) At $Q = 20$:

- $TC = 200 + 200 + 200 = \$600$
- $AVC = 10 + 10 = \$20$
- $ATC = 600/20 = \$30$
- $MC = 10 + 20 = \$30$

(c) Set the derivative of ATC to zero:

$$-\frac{200}{Q^2} + 0.5 = 0 \quad \Rightarrow \quad Q^2 = 400 \quad \Rightarrow \quad Q = 20$$

Minimum ATC $= \frac{200}{20} + 10 + 10 = \$30$. This matches the point where $MC = ATC$. In the table, ATC at $Q = 20$ is the lowest value.

Question 12. A firm's long-run average total cost is \$50 at 100 units, \$40 at 200 units, \$40 at 400 units, and \$52 at 800 units.

(a) Over what output range does the firm experience economies of scale, constant returns, and diseconomies of scale?
(b) What is the minimum efficient scale?

Answer Key

(a) Economies of scale: 100 to 200 units (LRATC falls from \$50 to \$40). Constant returns: 200 to 400 units (LRATC flat at \$40). Diseconomies of scale: beyond 400 units (LRATC rises to \$52 at 800 units).

(b) The minimum efficient scale is 200 units, the smallest output at which LRATC reaches its minimum of \$40.

Question 13. A firm has the short-run total cost

$$TC = 100 + 4Q + Q^2, \qquad MC = 4 + 2Q$$

(a) If the market price is \$20, how many units should the firm produce, and what is its profit?
(b) If the market price falls to \$10, should the firm produce or shut down? Explain using the loss from each choice.

Answer Key

(a) Set $P = MC$: $20 = 4 + 2Q$, so $Q = 8$.

- $TR = 20 \times 8 = \$160$
- $TC = 100 + 32 + 64 = \$196$
- Profit $= 160 - 196 = -\$\mathbf{36}$

The firm operates because price (\$20) covers AVC at $Q = 8$ (AVC $= 4 + 8 = \$12$). Shutting down would lose the full \$100 of fixed cost, so producing (loss of \$36) is better.

(b) Set $P = MC$: $10 = 4 + 2Q$, so $Q = 3$.

- $TR = 10 \times 3 = \$30$
- $TC = 100 + 12 + 9 = \$121$
- Profit $= 30 - 121 = -\$\mathbf{91}$

Price (\$10) is above AVC at $Q = 3$ (AVC $= 4 + 3 = \$7$), so the firm still covers its variable cost and produces. Its loss of \$91 is less than the \$100 loss from shutting down, so the firm should keep producing in the short run.

A

A.1 Main Summary